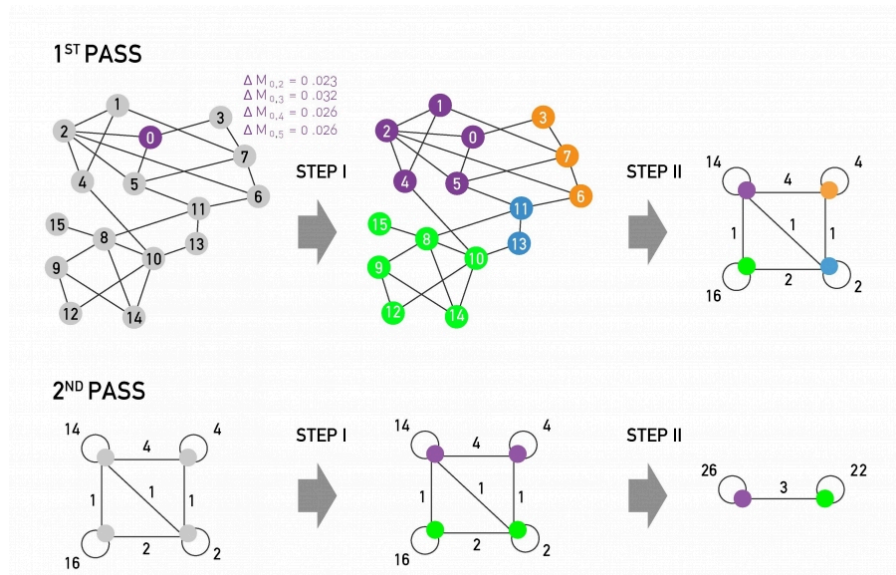


L7: Louvain Algorithm

The Louvain algorithm is a more computationally efficient modularity maximization algorithm than the previous greedy algorithm.

Even if the original network is unweighted, the Louvain algorithm creates a weighted network (*as described later*), and for this reason, all modularity calculations are performed using the modularity formula for weighted networks we saw earlier.



The main steps of the Louvain algorithm. Each pass consists of two distinct steps. The sum of Steps I & II are called a pass. The network obtained after each pass is processed again, until no further increase of modularity is possible. (Modified from Image 9.37 from networksciencebook.com)

The Louvain consists of two steps:

Step-I

Start with a network of N nodes, initially assigning each node to a different community. For each node i , we evaluate the modularity change if we place node i in the community of any of its neighbors j .

We move node i in the neighboring community for which the modularity difference is the largest -- but only if this difference is positive. If we cannot increase the modularity by moving i , then that node stays in its original community.

This process is applied to every node until no further improvement can be achieved, completing Step-I.

The modularity change ΔM that results by merging a single node i with the community A of a neighboring node is similar to the formula we derived earlier – but for weighted networks:

$$\Delta M_{i,A} = \left(\frac{W_{A,int} + 2W_{i,A}}{2W} - \left(\frac{W_A + W_i}{2W} \right)^2 \right) - \left(\frac{W_{A,int}}{2W} - \left(\frac{W_A}{2W} \right)^2 - \left(\frac{W_i}{2W} \right)^2 \right)$$

where W_A is the sum of weights of all links in A , $W_{A,int}$ is the sum of weights of all internal links in A , W_i is the sum of weights of all links of node i ,

$W_{i,A}$ is the sum of weights of all links between node i and any node in A , and W is the sum of weights of all links in the network.

Step-II

We now construct a new network (*a weighted network*) whose nodes are the communities identified during Step-I. The weight

of the link between two nodes in this network is the sum of weights of all links between the nodes in the corresponding two communities. Links between nodes of the same community lead to weighted self-loops.

Once Step-II is completed, we have completed the first pass of the algorithm.

Steps I – II can be repeated in successive passes. The number of communities decreases with each pass. The passes are repeated until there are no more changes and maximum modularity is attained.

The visualization shows the expected modularity change $\Delta M_{0,i}$ for node 0. Accordingly, node 0 will join node 3, as the modularity change for this move is the largest, being $\Delta M_{0,3}=0.032$.

This process is repeated for each node, the node colors corresponding to the resulting communities, concluding Step-I.

In Step-II, the communities obtained in Step-I are aggregated, building a new network of communities. Nodes belonging to the same community are merged into a single node, as shown on the top right.

This process generates self-loops, corresponding to links between nodes in the same community that is now merged into a single node.

Food For Thought

The description of the Louvain algorithm here is only at a high-level. We recommend you also read the original publication that proposed this algorithm. Also, show that the computational complexity of this algorithm is $O(L)$.